

# The Quest For QUANTUMLAND



**It's been a few more years now; so are we any closer to the age of quantum computing?**

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*"He who tries to understand quantum theory vanishes into a black hole, never to be seen again."*  
—Richard Feynman  
Celebrity Physicist

Language is poor indeed when it comes to certain things. How does one describe the state of mind of someone reading up on quantum theory? Baffled. Boggled. Stupefied. Dumbstruck. Confused. On the list goes—without capturing the essence of the feeling one gets when trying to grasp that ultimate theory of physics, which describes all of existence.

After all, what theory tells you that something can be a wave and a particle at the same time? That a cat in a chamber can be dead and alive at the same time—until the chamber is opened? That there is *only one electron*, which manifests itself as many electrons in the dance that is the cosmos?

As if that were not enough, we have the multiverse. You guessed it, "multiverse" is short for "multiple universes": not a new concept, but one that is shoved in our faces with renewed vigour when we talk of *quantum computing*. You see, several scientists believe there are a multitude of universes, each identical to the other until a bifurcation point happens. Such a point comes about

when a decision is made. To illustrate, say you're in a restaurant, and the waiter asks you "Tea or coffee sir?" Now, you have a choice, and decide on either tea or coffee—this is a bifurcation point, where the universe splits in two. In one of them, you're sipping tea, and in the other, coffee. These universes are only slightly different from each other—until more bifurcation points come along, and so on. That means infinities upon infinities of universes, and more interestingly, you are actually a billionaire with a harem in *some* universe!

## "Shadow Photons"

But what's the point of considering other universes when we can't interact with them? Ah, but we can. Let's look at a simple enough experiment: light passing through two slits in an opaque barrier and forming a pattern on a screen beyond that. (*Figure 1 on the next page.*) We'll leave out the technicalities; when there are four slits, the pattern changes to what's seen in Figure 2! The light, as you know, consists of photons. So *what* is blocking the photons such that the dark regions appear in Figure 2? It's photons from the other universes, and this is what is usually called *interference*. These "other" photons are absolutely undetectable in "this" universe, and show themselves as existing only when they interfere with "real" photons. To repeat, that's the only way we know they exist, and that's the only way we can tell the existence of the other universes!

That is justification enough for the existence of other universes, for two reasons: one, this is a great way of explaining how photons interfere, and two, lack of space here doesn't permit further discussion!

Illustration Pradip Ingale